Two-Dimensional Signal Processing: Continuous-Space

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Key Concepts

1) Two-dimensional (2D)signal processing is used for analysis of images.

2) Understanding two-dimensional signal processing naturally leads to extensions to three dimensions for volumetric imaging like MRI (magnetic resonance imaging) and CT (computed tomography)

3) 2Dsignal processing over space is a generalization of one-dimensional signal processing over time.

a) Linear shift invariant (LSI) systems extend the notion of linear time invariant systems.

   i) Linear means that the output of the system when the input is a weighted sum of signals is a weighted sum of the individual signal outputs.

   ii) Shift invariant means that the system behaves the same no matter where the image is centered.

   iii) The output of an LSI system is the two-dimensional convolution of the input and the system impulse response.

   iv) The 2D impulse response is known as a point-spread function.

b) The 2D Fourier transform extends the one-dimensional case by having two frequency variables.

c) If a 2D sinusoid is applied to an LSI system, the output is also a 2D sinusoid of the same frequencies. The system only modifies the amplitude and phase of the input.

d) The 2D Fourier transform converts 2D convolution into multiplication of 2D Fourier transforms. This allows us to define the frequency response of a 2D LSI system analogous to the frequency response of a one-dimensional LTI system.

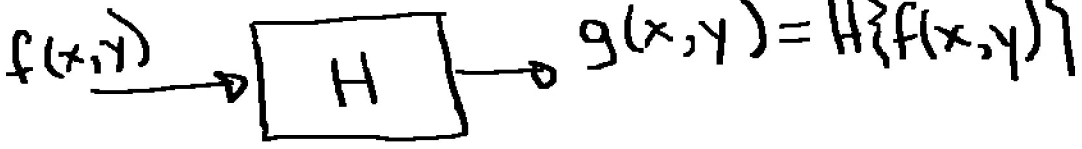
2-D Signal Processing: Continuous Space

- Imaging
- Extension to 3+D



$f(x,y)$

2-D Systems



Linear, Shift Invariant (LSI) Systems

1) $H\{\alpha_1 f_1(x,y) + \alpha_2 f_2(x,y)\} = \alpha_1 H\{f_1(x,y)\} + \alpha_2 H\{f_2(x,y)\}$

2) $H\{f(x-x_0, y-y_0)\} = g(x-x_0, y-y_0)$ where $H\{f(x,y)\} = g(x,y)$

LSI Input-Output : 2-D Convolution

$$g(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x-\alpha, y-\beta)\, f(\alpha,\beta)\, d\alpha\, d\beta$$

where $h(x,y) = H\{\delta(x,y)\}$ - impulse response
- point spread function

2-D Fourier Transform

2-D FT

$$F(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\, e^{-jux}\, e^{-jvy}\, dx\, dy$$

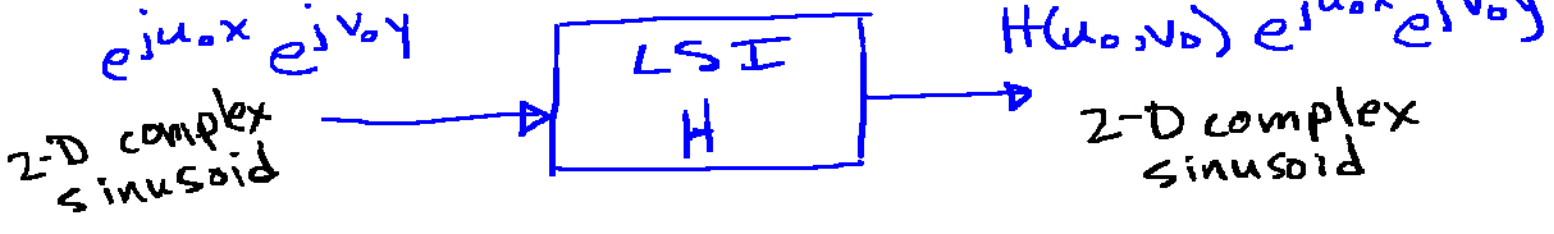
$$= \int_{-\infty}^{\infty} \underbrace{\left[\int_{-\infty}^{\infty} f(x,y) e^{-jux}\, dx\right]}_{\text{FT over } x} \underbrace{e^{-jvy}\, dy}_{\text{FT over } y}$$

u: freq vbl in x
v: freq vbl in y

2-D Inv FT

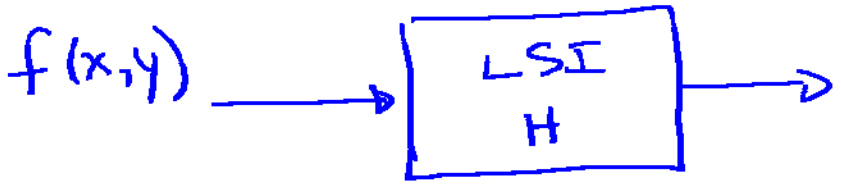
$$f(x,y) = \frac{1}{(2\pi)^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(u,v)\, e^{jux}\, e^{jvy}\, dx\, dy$$

Convolution - Multiplication Property



$$H(u_0,v_0) = H(u,v)\Big|_{\substack{u=u_0\\v=v_0}}$$

and $H(u,v) = FT\{h(x,y)\}$



$$g(x,y) = h(x,y) * f(x,y)$$

$\updownarrow$ FT

$$G(u,v) = H(u,v)\,F(u,v)$$